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Start-ups survival through a crisis. Combining machine learning with econometrics to measure innovation

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ABSTRACT

This paper shows how data science can contribute to improving empirical research in economics by leveraging on large datasets and extracting information otherwise unsuitable for a traditional econometric approach. As a test-bed for our framework, machine learning algorithms allow to create a new holistic measure of innovation following a 2012 Italian Law aimed at boosting new high-tech firms. We adopt this measure to analyse the impact of innovativeness on a large population of Italian firms which entered the market at the beginning of the 2008 global crisis. The methodological contribution is organised in different steps. First, we train seven supervised learning algorithms to recognise innovative firms on 2013 firmographics data and select a combination of those models with the best prediction power. Second, we apply the latter on the 2008 dataset and predict which firms would have been labelled as innovative according to the definition of the 2012 law. Finally, we adopt this new indicator as the regressor in a survival model to explain firms' ability to remain in the market after 2008. The results suggest that innovative firms are more likely to survive than the rest of the sample, but the survival premium is likely to depend on location.

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1. Introduction

This paper shows how data science can contribute to empirical research in economics by using large datasets and extracting information otherwise unsuitable for a traditional econometric approach. However, research questions drawn from economic theory and the assumptions behind the econometric modelling have guided our choices in benefiting from the richness of the data science. As an exemplary case and further contribution to the literature, we have applied this framework to evaluate the performance and survival rate of innovative start-ups (hereafter, INNs) *vis-à-vis* other types of newly founded firms (non-innovative start-ups, hereafter NOINNs) for which empirical the evidence shows contrasting results.

Greater consensus can be found around the two major challenges that undermine a robust causal relation between innovation and survival probability. First, most commonly selected proxies and measures for innovation have revealed serious limitations in capturing innovation (OECD 2018). Second, firm survival may depend on many internal and external factors, and therefore the

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innovation effect is not easy to isolate and might suffer from confounding issues (Freeman 1994). Nevertheless, this paper does not want to be merely another study on the innovation effect on firm survival. Our contribution is primarily methodological. We adopt an alternative and holistic-like measure of innovation drawn from the Italian national regulation.

We analyse the effect of innovation on the survival probability of a large sample of Italian start-ups established in 2008, the very first year of the economic and financial crisis that marked a strong acceleration in the decline of the Italian industrial sector. Assuming that the crisis exacerbated both market risks and financial constraints, this database offers an extraordinary opportunity to test the effect of a very strong selection mechanism. If there is any truth into an evolutionary framework, which describes industrial dynamics as triggered by the evolutionary mechanism of entry and selection, we should be able to observe it in a time of crisis.

Our empirical strategy is able to effectively relax some of the constraints imposed by the traditional inferential analysis by integrating a data science approach with econometrics according to three steps.

First, we adopt a definition of ‘innovative start-up’, built on the multiple criteria prescribed by the 2012 Italian regulation aimed at boosting new high-tech firms through a programme of incentives. Therefore, we extract all available new entrant firms in 2013 from AIDA, the Bureau Van Dijk database, including start-ups both registered and not registered as innovative according to the Italian law. After a data cleansing process, we implement a supervised machine learning approach based on the training of seven algorithms (namely recursive partitioning, classification and regression trees, Logit regression, bagging, naïve Bayesian classifier, and artificial neural network) to predict the probability of being classified as INNs using 124 firmographics variables. Since the literature on innovation considers sectors and locations as important confounding effects in explaining the survival probability of firms, we have excluded them from the training-set of the machine learning algorithms. This will allow us to eventually include these variables in an econometric framework, without the risk of describing spurious relationships.

Second, from the same database, we extracted a sample of new firms which entered the market in 2008 and which faced a highly competitive environment due to the economic and financial crisis. We selected a combination of the above algorithms to predict the probability of being INNs.

Third, once we discriminated between INNs and NOINNs, according to the above multicriteria definition, we estimated the survival probability of firms over a ten years period (2008–2018), with a Cox proportional hazards model including sector and location effects. Without the use of machine learning algorithms, this innovative measure of innovation could not have been created and, without a clear theoretical input from the literature and the econometric assumptions to guide the machine learning modelling, this new indicator would have been useless.

In a nutshell, the contribution of this paper is threefold:

- we provide new evidence on the survival of innovative start-ups;
- we design and calculate a new measure to detect innovative firms, which goes beyond the question of survival probability;
- we develop an original methodological framework to combine data science and econometrics.

The paper proceeds as follows. In Section 2, we present our methodological approach and explain how it can contribute to economic empirical analyses in general, while Section 3 positions our contribution in the debate around the role of innovation in fostering survival probability, as a specific case to test our methodology. Thereafter, in Section 4, we present the machine learning process which leads to the creation of the new indicator for innovation. In Section 5, we carry out a survival analysis, and, finally, in Section 6 we summarise and discuss the main results of the paper, as well as new challenges.

2. Data science: an opportunity for the creation of new variables

The data science paradigm is the result of the recombination and convergence of a few complementary technologies, which extract information and knowledge from a very large dataset: algorithms, computational power, collection and storage of digitised data (Estolatan et al. 2018). Along with Varian (2014), this change of paradigm has provided economists with an expanded set of analytical tools to explore data and acquire information. In particular, we can recognise at least three types of approaches to data analysis, which differ widely among each other in both their goals and tests for uncertainty.

Econometrics is the most popular and oldest set of statistical methods aimed at highlighting causal relations between economic variables. The external validity of its results relies on statistical inference, which requires available observations to be a random sample of the population. Well-known techniques have been developed for non-random data or for the generation of truly random data in experimental settings. Nevertheless, estimator properties have been derived on a limited class of mostly linear models with several statistical limits. Feedback between variables is difficult to handle and even prohibited between dependent and independent variables; the presence of heteroskedasticity and autocorrelation in the error terms of the econometric model need to be carefully addressed; and an excess of multicollinearity between covariates raises serious computational issues.

On the whole, the capability to highlight statistically robust causal relations creates a heavy constraint on the variety of models that can be implemented. This limitation impinges on the explanatory power and the performance of out-of-sample predictions. Moreover, the complex reality represented by big data rarely fits into the required econometric assumptions, nor does the data collection always occur in controlled settings. For this reason, econometrics lacks the ability to fully exploit big data.

Data science provides empirical researchers with a variety of supervised machine learning algorithms (see, among others, Kotsiantis, Zaharakis, and Pintelas 2007) that are quite suitable for prediction and/or classification. Technically, predictive modelling is the problem of approximating a mapping function (f) given input data (X) to predict an output value (y). In this framework, algorithms are trained on large numbers of cases and variables (training-set) and learn from a target category to assign new observations. External validity, i.e. the variance of the estimates in out-of-sample predictions, is tested on a partition of the available data (test-set), which is not employed in the learning process: namely, for algorithm prediction of an unobserved category. For this reason and contrary to the econometric approach, a machine learning algorithm is not restricted by any assumption and the only objective function is to maximise the prediction power on the test-set. A clear trade-off emerges between the adoption of models aimed at finding causal relations and models aimed at predicting or classifying a phenomenon (Shmueli 2010). The former are cautious in the data selection and need to be relatively simple in the functional form to approximate data points and to minimise the mean square error of estimators in order to confirm the underlying theory. Extremely simple models tend not to fit data well enough (under-fitting) and their explanatory power remains limited to the few variables involved, which are not necessarily those which explain the total variability of the phenomenon outside the sample. In other words, they might be unable to account for the complex nature of social phenomena like innovation characterised by the interdependence and interaction of a variety of agents and factors (Antonelli 2009; Fontana 2010). The latter are meant to obtain excellent prediction performance, but can become blind to any causal relation and risks capturing the background noise of the data.¹

A second approach, based on machine learning, takes the form of unsupervised algorithms which create a data partition without any *a priori* restriction on the number and type of categories to be generated. Clustering algorithms (Macqueen 1967), self-organising maps (Carlei and Nuccio 2014) and, more recently, topic modelling for text analysis of the economic literature (Ambrosino et al. 2018) belong to this group. In unsupervised algorithms, the model validation is pursued by an *ex-post* educated interpretation of the result.

We claim that research in economics can take advantage of the latter approaches. Even though someone has envisioned the end of economic theory and the transition towards a purely data-driven type of science (Anderson 2008; Prensky 2009), other authors suggest that the large availability of data, which reveals the complexity of the relationships in the observed reality, actually calls for more theory (Kitchin 2014; Ambrosino et al. 2018; Nuccio and Guerzoni 2019; Carota, Durio, and Guerzoni 2014; Gould 1981). Data and its analysis can still act as a powerful hypothesis-mining engine (Jordan 1998; Carota, Durio, and Guerzoni 2014) and provide new theoretical ideas, which then need to be filtered through a theoretical framework. Unfortunately, the implementation of machine learning within economics is still not widespread, except for a few methodological contributions (Athey 2018; Varian 2014).

The methodological conceptualisation behind our empirical exercise shows that economic theory, econometrics and data science can be strongly complementary (Figure 1). Even within a traditional framework of economic theory and hypotheses testing, the large availability of data can be exploited to create a new independent (or dependent) variable which fits into a standard regression analysis. The theoretical input suggests that the hypotheses should be tested in a suitable econometric model, under specific assumptions about the variable distributions and their relations with other variables, to avoid issues such as multicollinearity or endogeneity. However, once these boundaries are set, data science can employ its brute force, prediction capability, and summarising potential, to extract information that could not have been otherwise used in a regression.

The test-bed of our approach is rooted in the long-standing controversial evidence on the different survival rates of INNs and NOINNs, on the extent to which this relation is distorted both by the type of indicator used to proxy innovation and a failure to control possible confounding variables like sector and location. Following the broad literature on this topic, we further argued that the existence of a survival premium for INNs can be tested best for the 2008 economic crisis, since the market selection mechanisms were more effective. Next, we turned to data science. We collected data on new firms established in 2013 when a new Italian Law, enacted on 17 December 2012, created incentives for start-ups identified as innovative firms. We employed a supervised machine learning approach to estimate the probability of the new 2013 firms belonging to the given class of 'innovative start-ups'. Then, we applied the same algorithm to predict which firms in 2008 could have been labelled as innovative, according to the 2012 law. As explained earlier, we partitioned the 2013 start-up sample in a training and a test-set. We applied a series of algorithms (see Appendix A.1) to the training set, with different degrees of complexity and with the aim of maximising their prediction power on the test-set. The algorithm, or the combination of algorithms, with

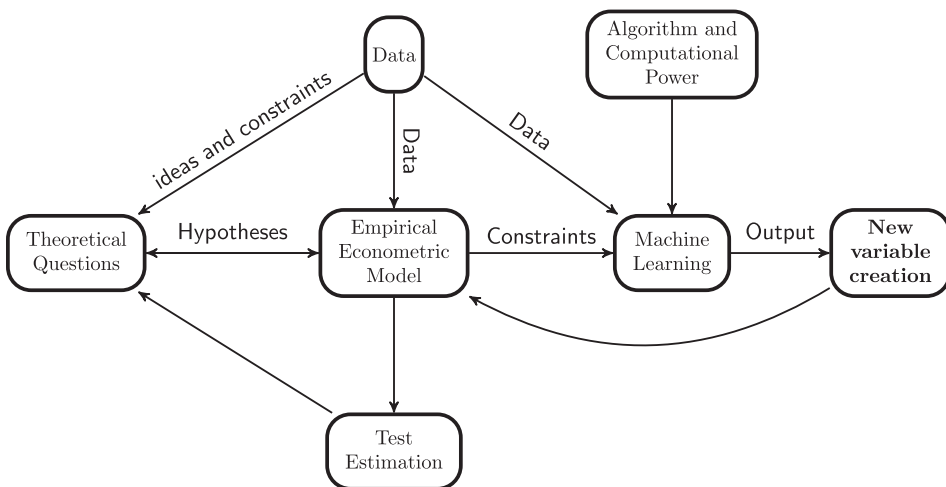


Figure 1. Data science for economics: the creation of a new variable.

Table 1. Summary of the methodology based on econometrics and machine learning tools.

Aim	Tool	Data	Process	Output
Measuring innovation	Supervised machine learning	Data on new Italian firms in 2013 by AIDA Bureau van Dijk	In the data, firms are flagged as innovative if they are registered as INNs according to the 2012 Italian Law 221/2012. The flag is used as a label to train a battery of models on a subset of the data. A second subset is used as a test-set to evaluate model performance	Model trained to detect INNs
Controlling for business cycle	Prediction	Data on new Italian firms in 2008 by AIDA Bureau van Dijk	The supervised algorithm, output of the previous line, is used on 2008 data to detect innovative firms	Classification of 2008 Italian innovative and non-innovative start-up
Controlling for sector and location	Multivariate survival analysis	Data on a 10-year panel of new Italian firms in 2008 by AIDA Bureau van Dijk, enriched by the classification created with the model	We run Cox regressions using the classification of the previous output as the main independent variable. Sector, location and interaction terms are added as controls	Estimation of the effect of being an innovative, sector and location on the survival probability

the best performance was then used to predict which firms from the 2008 sample could qualify as INNs. However, in order to include the new measure of innovation in an econometric model, the predictive algorithm was trained on all the available variables except for sector and location, which were used as covariates. The remainder of the paper discusses the details of this process, which is also briefly summarised in Table 1.

3. Innovation and survival

A key empirical stylised fact in industrial dynamics is the widespread heterogeneity of firms, ranging from the firms' size, productivity, rate of growth, and chance of survival (Griliches and Mairesse 1995). While these stylised facts clash with the mainstream economics narrative of the representative agent, it is fully aligned with an evolutionary economics framework according to which the mechanism of selection among heterogeneous agents explains the development of industries. In recent decades, great attention has been paid to the entry of new firms and how selection shapes their chance of survival and subsequent growth. More specifically, scholars focus on the role of the innovativeness of new firms and whether innovation can explain, or at least improve, their fit in the evolutionary landscape to improve/not improve their chance of survival.

This literature is rooted in the Schumpeterian idea of creative destruction generated by the entry of new firms (Schumpeter 1912, 1942), which are more prone to exploit both market and technological opportunities than a larger incumbent, since they are not locked into partly obsolete competencies (Malerba and Orsenigo 1997; Breschi, Malerba, and Orsenigo 2000). For this reason, we mainly observe successful INNs in highly technological sectors and with low cumulative knowledge; that is, previous competencies are not a strategic asset, but more likely a burden which hinders the possibility of exploiting new ideas (ibid.). This Schumpeterian view is shaped by the mythological idea of the entrepreneur as one who challenges the odds of fortune and after many attempts, eventually succeeds. When new innovative firms succeed, they can run as fast as gazelles (Acs and Mueller 2008) and, in a few cases, they can become rare unicorns (Simon 2016).

Beyond this narrative, innovative ventures come along with both advantages and disadvantages. Innovative firms can improve users and consumers utility by introducing better products and services (Guerzoni 2010); they are less myopic and can focus on emerging markets (Bower and Christensen 1996); they have less cognitive biases generated by previous activities (Aestebro, Jeffrey, and Adomdza 2007); and they are more dynamic (Teece 2012). At the same time, a high degree of

uncertainty might undermine their innovative efforts and lead them quickly toward closure. New markets are often characterised by high levels of uncertainty regarding consumer preferences (Guerzoni 2010) and the future development of technology (Dosi 1982). Further uncertainty can also be due to path-dependency, which might impede the diffusion of novelty (Dosi 1988), or to fierce competition, since other firms might take the lead in new markets (Fudenberg et al. 1983, among others). Finally, it might be more difficult to attract investments (Stucki 2013).

Although scholars have increasingly agreed upon an overall survival premium for innovative firms, this consensus does not seem rooted in strong empirical evidence. In reviewing the most relevant research on the topics (see, e.g. Arrighetti and Vivarelli 1999; Audretsch 1995; Calvo 2006; Cefis and Marsili 2005, 2006, 2012; Helmers and Rogers 2010; Santarelli and Vivarelli 2007; Wagner and Cockburn 2010), Hyytinen, Pajarinen, and Rouvinen (2015) find that the large majority of empirical works measuring the effect of innovativeness on survival probability accounts for a positive impact. A rigorous reading of the paper shows, however, that the evidence of positive effects is rather weak. For instance, Cefis and Marsili (2005) find a close to zero effect, while Cefis and Marsili (2006), although reporting a more robust result, do not control for a sector covariate. In a very detailed work, Colombelli, Krafft, and Quatraro (2013) show that the Kaplan-Meier survival function is virtually the same for innovators and non-innovators, while, on a sample of French start-ups, Colombelli, Krafft, and Vivarelli (2016) prove that being innovative is not enough for having better survival chances than non-innovative firms, in fact, the very small impact on survival emerges only for process innovation. Helmers and Rogers (2010) use patent activity as a proxy for innovation and find a mildly positive and significant effect of patenting on survival, but, owing to the large sample, the simple use of the p -value cannot really highlight anything conclusive. Indeed, when they repeat the analysis at the industry level and, thus, with less observations for regression, the coefficients are still negligible in size and the p -value is statistically significant only for some sectors. On a sample of U.S. listed companies, Wagner and Cockburn (2010) find a very small impact of owning patents on firm survival and the coefficients are again mildly significant and only for a few specifications. The survey includes also papers like Boyer and Blazy (2014) which show a negative impact of innovativeness on survival. Therefore, we tend to disagree with Hyytinen, Pajarinen, and Rouvinen's (2015, 12) conclusion that '[...] [T]he prevailing view in the empirical literature appears to be that there is a positive association between the innovativeness of firms and their subsequent survival'. Furthermore, there is also a mild disagreement on the effect that different types of innovation might have. For instance, Cefis and Marsili (2019) show that non-technological innovation can be detrimental for survival, while Expósito and Sanchis-Llopis (2019) find that the non-technological innovation can improve business performance.

We highlight three main issues with the present empirical literature which might explain the disparate effect of innovation on the performance of entrants (Audretsch 1995).

3.1. *The measurement of innovation*

The research community of innovation studies has always acknowledged a number of shortcomings in the measurement of innovation, but this is rarely addressed in empirical works and mostly relegated to footnotes. Even the 2018 Oslo Manual (OECD 2018) devotes merely a few sentences on the limits of measuring innovation, which we will shall mention in the next few paragraphs. The proxies adopted in empirical research for measuring innovation can be roughly divided into two groups: proxies for innovation input and proxies for innovation output. The input of the innovation process are typically R&D investments and highly skilled labour, while the innovation output is usually the number of products, or process innovations, or patent applications. Figures on R&D expenses and personnel costs usually come from register data; patents are easily identified at patent offices, and extensive literature exists on their uses. Information on the number and nature of new products or processes can be found in self-reported surveys such as the Community Innovation Survey.

Each of these empirical proxies have important downsides, which are even more severe for recently established firms. R&D expenses in register data are not always representative of real R&D activity, especially in small companies, where R&D is not pursued in a formal way, or conversely, in high-tech start-ups where the R&D activity is spread out across the firm's operation. The sheer number of product and process innovations creates a bias toward misrepresenting the respondent's concept of innovation (OECD 2018). Moreover, surveys do not investigate all existing firms, especially small firms, which is what start-ups typically are, and, thus, the process of sample selection can induce bias, reduce the possibility of panel data, lessen the degrees of freedom of the model, and raise inference issues. As for patents, there is clear evidence on the extreme variance in the propensity to patent, both between and within sectors, since in many cases, especially for process innovation, the appropriability of the economic returns of intellectual property rights (IPR) can be achieved by means of secrecy (Harabi 1995). In addition, patents are an indicator of the inventing activity, and only rarely turn out to be commercially viable, since patenting is pursued for a vast array of purposes.²

These measurement issues are even more stringent for start-ups since the balance sheets in the first years are rarely an accurate representation of the firm's business, and, as for patents, start-ups could still have them in the application process, or decide not to patent since in some contexts time-to-market might be much more important than a strong IPR.

3.2. Business cycles as a confounding effect

New firms can prosper, or fail, for many reasons which do not necessarily relate to economic or technological conditions at the micro-level. For instance, vulnerable firms might survive in a growing economy even if they are not profitable, while selection mechanisms become stricter in downturns. The literature on the economic and financial crisis agrees that recessions usually hinder the survival of existing firms. Peric and Vitezic (2016) review the literature on the adverse effects of crises on existing firms and highlight their main aspects such as: production and product lines (Liu 2009); sales (Cowling et al. 2014); employment (Rafferty et al. 2013); investments (Campello et al. 2011; Bucă and Vermeulen 2017); performance (Akbar and Ormrod 2013); risk tolerance (Hoffmann, Post, and Pennings 2013; Inklaar and Yang 2012); and business confidence (Zenghelis 2012; Geels 2013; Peric and Vitezic 2016, 3). However, other entrepreneurial studies have stressed the positive effects that can be produced by an economic crisis (Bartlett 2008). This is especially true for those firms that can identify market changes and react promptly to exploit newly created opportunities (Hodorogel 2009). For this reason, difference in the survival rate between innovative or non-innovative firms should be spotted more precisely from the cohort of firms born in 2008, when business constraints were already more binding.

3.3. Sectors and location as confounding effects

Since Pavitt (1984), sector specificity has been widely acknowledged to play a crucial role in explaining the performance of a sector, especially in terms of innovation. Along the same lines, Malerba and Orsenigo (1997) developed a theory and provided strong empirical evidence showing that the technological base, underlying the activities of a sector, is a key driver of the innovative performance of firms. Sectors characterised by high technological opportunities, low appropriability, and a low cumulativeness of technological knowledge, experience a high entry rate of innovative firms, but also a high exit rate. On this premise, Malerba et al. (1999) introduced the idea that the sector is the proper unit of analysis for the development of models able to replicate the history of industrial dynamics in the computer industry. Sector based history-friendly models have thereafter been applied to various industries: such as pharmaceutical (Malerba and Orsenigo 2002), textile (Fontana, Guerzoni, and Nuvolari 2008), and DRAM (Kim and Lee 2003).³

Along the same lines, the industry life cycle approach theorised and showed that the early stages of new industries attract most of the entries, but at the same time experience the highest rates of

failures (Klepper 1996; Geroski 1995). Within an evolutionary perspective, this can be framed as the costly process of trial and error at the industry level in which many enter, but most do not survive. The firms that survive, thereafter, exhibit a more than proportionate growth rate based on their performance (*ibid.*). Thus, Pavitt (1984) taxonomy, Malerba and Orsenigos (1997) classification, and Kleppers (1996) industry life cycle approach suggest that a bias exists for innovative firms in specific sectors, and the survival rates might differ between innovative and non-innovative firms because of the natural self-selection of innovative firms in explicit sectors with specific survival patterns.

Similarly, since the distribution of economic activities is very uneven, region-specific fixed effects can introduce a further confounding effect when analysing the survival rate of firms. The impact of a region on economic performance is heavily determined by the spatial distribution of economic activity at the industry level. However, Acs, Armington, and Zhang (2007) show that, even after controlling for both the industry mix of an area and its degree of specialisation, location continues to impact the survival probability. Indeed, Sternberg and Litzenberger (2004) and Sternberg (2009) recall and demonstrate that entrepreneurship is mainly 'a regional event' (Feldman 2001) for many reasons, which can be broadly defined as agglomeration economies (Leone and Struyk 1976; Sorenson and Audia 2000) of the regional system of innovation (Howells 1999), which includes, among other things: local government policies, specific user-producer interactions (Rothwell 1994), the presence of an entrepreneurial atmosphere (Ciccone and Hall 1996), the role of cities (Lee, Florida, and Acs 2004) and industrial clusters (Rocha 2004), and the presence of higher education institutions or research centres (Fetters, Greene, and Rice 2010). Knowledge spillovers are the key input in the complex process of innovation, especially for new entrants (Audretsch and Feldman 1996, 2004). Nevertheless, there is no consistent use of controls for industries and regions that the theoretical literature has suggested as being the most important. For instance, none of the works discussed (Arrighetti and Vivarelli 1999; Audretsch 1995; Calvo 2006; Cefis and Marsili 2005, 2006; Colombelli, Krafft, and Vivarelli 2016; Helmers and Rogers 2010; Santarelli and Vivarelli 2007; Wagner and Cockburn 2010) control for location, nor do all of them control for the sector impact.

Both theoretical and empirical considerations trigger the necessity for a novel approach to the problem of survival. This paper provides a solution to the three issues discussed above and provides evidence of different survival rates for INNs and NOINNs by: (i) clearly introducing a new broad measure of innovativeness; (ii) focusing on the total number of new Italian firms in times of economic crisis; and (iii) controlling for sectors and location, as suggested by the theory.

4. Data and methodology

The AIDA (Analisi Informatizzata delle Aziende) database, provided by Bureau van Dijk, contains comprehensive information on all Italian-owned companies required to file their accounts, including whether they have registered as INNs, or not, according to the Italian law (Ministerial Decree 179/2012).⁴

Each firm in AIDA is described by 427 variables belonging to the following macro categories: (i) identification codes and vital statistics; (ii) activities and commodities sector; (iii) legal and commercial information; (iv) index, share, accounting, and financial data; (v) shareholders, managers, company participation. Only variables in category (iv) are observed for different years. In the construction of our sample, we excluded category (v), since the nature of this data is very specific to each start-up and not suitable for prediction analysis, nor for econometrics. Despite its considerable size (Table 2), the AIDA database does not cover the entire range of Italian firms as does the Italian Board of Trade (IBT) database; for instance, amongst others, banks, insurance companies, and public entities are not included. Still, our sample varies from a minimum of 62,934 observations in 2009 (about 21.8% of new firms) to a maximum of 74,508 in 2010 (about 28% of new firms). The dataset is restricted to 276 variables for all firms entering the Italian market from 2008 to 2015 and 262 variables are observed from the starting year of activity until 2015. For new Italian firms established in 2008, we have a balanced panel with ten selected variables up to 2018. Here we

Table 2. INNs and NOINNs in the collected sample according to their initial year of activity.

	2008	2009	2010	2011	2012	2013
INNs	0	4	51	320	531	1010
NOINNs	65,088	62,930	74,457	71,599	65,653	67,306
Total	65,088	62,934	74,508	71,919	66,184	68,316
% Italian Start-ups (IBT)	22.7%	21.8%	28.1%	27.2%	24.0%	24.7%
% INNs (AIDA)	0%	0.01%	0.07%	0.4%	0.8%	1.5%
% INNs in AIDA/ INNs in IBT	0%	21.05%	43.22%	124.51%	112.03%	100.50%

Note: the value can exceed 100% for a different account of firms which ceased to exist.

Source: AIDA and IBT.

focus on two sets of firms, which entered the market in 2008 and 2013, respectively. Since not all information is mandatory for each category of firms, the dataset is characterised by many missing values. Therefore, we conducted a careful missing value analysis, which brought us to exclude certain variables and observations and obtain two samples of firms (45,576 (2013) and 39,295 (2008)). Appendix 1 includes details of our cleansing methodology.

4.1. Measuring innovation: innovative start-ups and Legislative Decree 221/2012

In the previous section, we suggested that one cannot rule out the possibility that the weak evidence, provided in the empirical literature on the survival rate of firms and their level of innovation, depends on measurement issues. Usually, the literature assesses the innovativeness of firms looking either at the inputs of the innovation process, such as R&D expenses, or the number of employed researchers, or the outputs of the process, such as the patent pool or the number of innovations. For this reason, we adopt a new definition of ‘innovative start-ups’ which allows several aspects of their innovative activity to be grasped simultaneously.

Starting from 2013, a new Italian class of firms named ‘innovative start-ups’ has been identified through Legislative Decree 221/2012. The policy enforced by this law encourages the creation of companies with specific characteristics, defines the level of incentives, and sets up a dedicated section in the Italian companies register.⁵ Start-ups applying for incentives must meet the following conditions:

- be new or have been operational for less than five years;
- have their headquarters in Italy or in another EU country, but with at least one production site in Italy;
- have a yearly turnover lower than 5 million euros;
- not distribute profits;
- produce, develop and commercialise innovative goods or services of high technological value;
- not be the result of a merger, split, or selling-off of a company or branch;
- be innovative in character, which can be identified by at least one of the following criteria:
 - at least 15% of the company's expenses are attributed to R&D activities (satisfied by 64.97% of the INNs);
 - at least 1/3 of the total workforce are PhD students, holders of a PhD, or researchers; alternatively, 2/3 of the total workforce must have a Master's degree (satisfied by 29.68% of the INNs);
 - the enterprise is the holder, depositary or licensee of a registered patent (industrial property) or the owner of a programme for original registered computers.

According to the actual composition of the INNs, only 2.7% satisfied all of the requirements and 11.08% met two to three of the requirements. However, we do not know which specific criteria from AIDA were satisfied. We only have aggregate data from the IBT, presented in Table 3 for 2013.

Table 3. Number of 2013 INNs which satisfy the three possible requirements of Law 221/2012.

	First Req.		Second Req.		Third Req.	
No	360	(35.82%)	709	(70.55%)	787	(78.31%)
Yes	645	(64.18%)	296	(29.45%)	218	(21.70%)

Source: IBT, March 2016.

Legislative Decree 221/2012 provides us with a new tool to identify INNs with some advantages over previous indicators of innovativeness:

- it focuses on small firms, which are very likely to be truly new entities and not subsidiaries or foreign green-field entrants;
- all innovative firms are focused on innovative goods or services;
- they need to have at least one of the usual proxies for innovative input and output, but not necessarily a specific one, as in the other measures.

Table 2 shows the INNs number with respect to the total number of sampled firms and the percentage of firms in the data over the entire range (source: see Footnote 5) of new Italian firms. AIDA covers about one fifth of new Italian firms; the firms of the self-employed, professionals and other minor activities are not required to register their accounts. In 2013, around 1.5% of new firms were registered as innovative start-ups.

Table 4 describes the distribution of INNs across the ATECO2007 sector classification and shows that INNs are principally active in service and manufacturing (code J and C, respectively). The map of INNs in 2013 (see Table 5 and Figure 4 in Appendix 3) shows a striking concentration in two regions, Lombardy and Lazio, and their capital cities, Milan and Rome, which attract about one out of three INNs. Finally, we provide the main features of INNs by presenting a few summary tables on their state of activity after 2 years (98% were still active in 2015, see Table 6). It should be noted that in the survival analysis we consider firms as dead if they had a negative exits, such as closures or bankruptcy, mergers and takeovers are excluded.

Table 4. One digit ATECO2007 of the 2013 INNs ($n=1010$) and NOINNs ($n= 67,306$).

ATECO	NOINNs start-ups		INNs start-ups	
A	1039	(1.54%)	6	(0.59%)
B	46	(0.06%)	0	(0.00%)
C	7112	(10.56%)	161	(15.94%)
D	703	(0.10%)	10	(0.10%)
E	326	(0.48%)	4	(0.39%)
F	8290	(12.32%)	14	(1.39%)
G	15,415	(22.90%)	59	(5.84%)
H	2640	(3.92%)	3	(0.30%)
I	6072	(9.02%)	0	(0.00%)
J	3113	(4.63%)	431	(42.67%)
K	1309	(1.94%)	1	(0.10%)
L	3193	(4.74%)	0	(0.00%)
M	4963	(1.54%)	261	(25.84%)
N	4260	(7.37%)	37	(3.66%)
O	6	(0.00%)	0	(0.00%)
P	666	(0.99%)	7	(0.69%)
Q	1307	(1.94%)	6	(0.59%)
R	1865	(2.77%)	3	(0.30%)
S	1098	(1.63%)	4	(0.40%)
T	1	(0.00%)	0	(0.00%)
Total	67,306		1010	

Source: AIDA.

Table 5. Regional distribution (NUTS2) of the 2013 INNs ($n=1010$).

Italian region	Number of innovative start-up		Italian region	Number of innovative start-up	
Abruzzo	16	(1.58%)	Molise	0	(0.00%)
Basilicata	4	(0.40%)	Piemonte (Torino - 57)	72	(7.13%)
Calabria	14	(1.39%)	Puglia	51	(5.05%)
Campania	65	(6.44%)	Sardegna	30	(2.97%)
Emilia-Romagna	111	(10.99%)	Sicilia	39	(3.86%)
Friuli-Venezia Giulia	26	(2.57%)	Toscana	53	(5.25%)
Lazio (Roma - 102)	111	(10.99%)	Trentino-Alto Adige	27	(2.67%)
Liguria	15	(1.49%)	Umbria	13	(1.29%)
Lombardia (Milano - 155)	229	(22.67%)	Valle d'Aosta	3	(0.30%)
Marche	47	(4.65%)	Veneto	84	(8.32%)

Source: AIDA.

Table 6. State of the 2013 INNs after two years ($n=1010$).

Status	Number of INNs	
Active	994	(98.42%)
Close down	0	(0.00%)
Failed	1	(0.1%)
Liquidation	15	(1.49%)

Source: AIDA.

4.2. Isolating the innovators' premium from confounding effects

Given the current legislative situation, the introduction of the 2012 Legislative Decree, allows us to perform a survival analysis, distinguishing between innovative and non-innovative firms, from 2013 onwards (not for those created in 2008). In this subsection, we explain how a machine learning algorithm can be trained and tested on 2013 data to identify INNs created in 2008, based on a vast array of other firmographics. We only excluded from the analysis the industry sector and the geographical location that otherwise would have made the new indicator unsuitable as a regressor in a model in which those same variables appear as other covariates. We also discarded from the training-set the investments in R&D and IPR, so to use them later as robustness checks for the prediction of the 2008 sample, in which we do not observe any target variables, as explained in Section 4.2.2. Finally, we focused on 2013 data since this is the closest available year to 2008 with the relevant information about the target variable (INNs). Predictive modelling learning from historical data is assumed to be static, but data evolves and must be analysed in near real time. The change over time of the statistical properties of the target variable, which the model is trying to predict, is also known as 'concept drift' (Liobait 2010). Therefore, to prevent deterioration of the prediction accuracy, one effective solution is to minimise the time interval between input data and prediction.

4.2.1. Training, test, and model selection to predict INNs

In this section, we apply different algorithms to classify firms as INNs and, thereafter, we compare their predictive power to select the most performing one. We have deployed seven widely used classifiers, which are described analytically in Appendix A.1:

- Recursive Partitioning (RPART);
- Classification Tree (TREE);
- Conditional Inference Tree (CTREE);
- Bagging (BAG);
- Logit Regression (LOGIT);
- Naïve Bayesian classifier (NB);
- Artificial Neural Network (ANN).

We train these algorithms on a 2013 random subset of 80% of the cleansed sample (36,401 firms including 563 INNs), and test them on the remaining 20% of the sample (9175 firms including 150 INNs). The dataset is unbalanced since the target variable (INNs) is underrepresented in the sample. The SMOTE algorithm (Chawla et al. 2002) is a well-known oversampling technique to address this problem because it artificially generates new examples of the minority class (here INNs) using the k -nearest neighbours of these cases. Furthermore, the majority class examples are also under-sampled, leading to a more balanced dataset. In due course, we summarily balance the data only in the training-set, while keeping the test-set unchanged, to evaluate the performance of factual data.

Each algorithm predicts the probability of a start-ups in the test-set, being an INN. The predicted probability, which ranges from 0 to 1, collapses to NOINN or INN according to a threshold (cut-off) chosen by the researcher on the basis of a model performance assessment. Our toolbox for comparing algorithms and selecting thresholds includes the analysis of receiver-operating characteristics (ROC) curves (Figure 2), and the density function of the predicted probability for both INNs and NOINNs. In detail, the ROC curves represent pairs of true positive and false positive rates of a classifier for a continuum of probability thresholds, and can be used to compare different classifiers. Specifically, the highest performing classifier is the one with the ROC curve closest to the upper-left corner (i.e. true positive rate close to 1 and false positive rate close to 0). If two classifiers are characterised by intersected ROC curves, it means that the two classifiers are better under different loss conditions.⁶ For each algorithm, we define the optimal cut-off as the one associated with the point which minimises the Euclidean distance between the ROC curve and the (0,1) point (see Appendix 2 for further details). Once a cut-off is set, confusion matrices (Table 7) summarise the number of correctly classified cases and the classification errors for each algorithm.

Also, the density function, for both NOINNs (or negative) and INNs (or positive), of the predicted probabilities (Figure 3) generated by the seven algorithms on the 2013 test-set can provide some insights on the model performance. In an ideal scenario represented in Figure 3(a), each distribution for the predicted probability of INNs and NOINNs is respectively skewed towards 1 and 0 and without a common support. Unfortunately, this is not actually the case; in most empirical analyses, I and II type misclassification errors can be relevant.

ROC curves offer an interesting comparative insight on the performance of the same model when used for estimating a causal relation instead of predicting a category, as discussed in Section 2. For example, the widely used Logit model is among the worse classifiers in terms of accuracy (Figure 3), confirming the theoretical framework developed in Section 2 and the main limitations of econometrics used to fit such types of data. Among the algorithms tested, BAG (with SMOTE) stands out

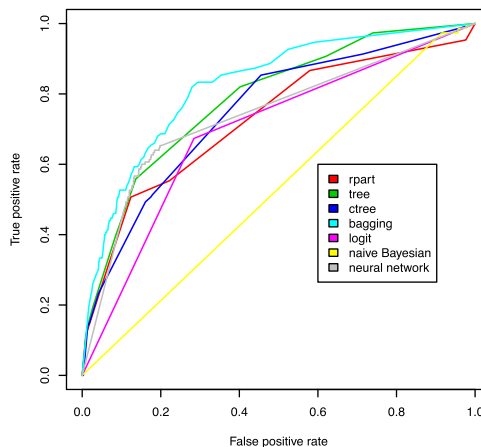
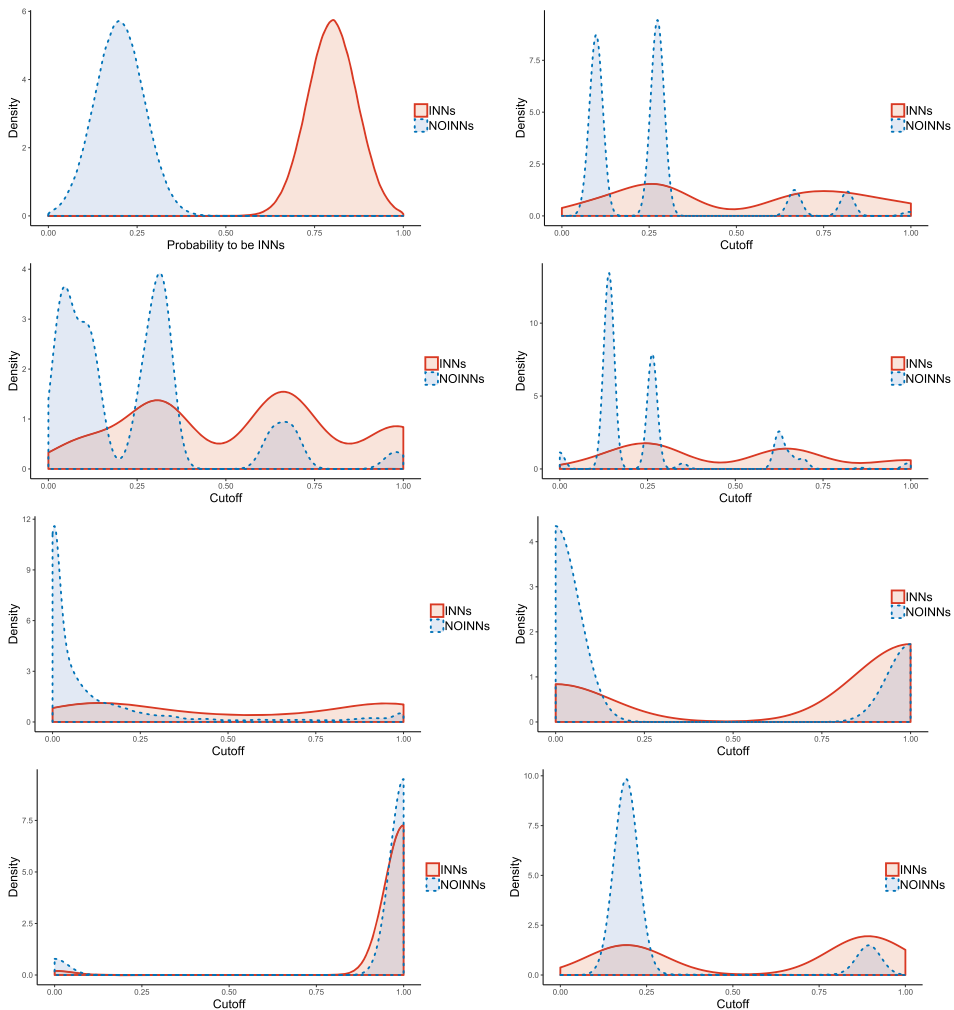


Figure 2. ROC curves on the 2013 sample with SMOTE.

Table 7. Confusion matrix of the seven algorithms with SMOTE (optimal cut-offs in parentheses).

Real data	RPART (0.2817)		TREE (0.3210)		CTREE (0.2632)		BAG (0.1200)		Total
	NOINNs	INNs	NOINNs	INNs	NOINNs	INNs	NOINNs	INNs	
NOINNs	7908	1117	7779	1246	6857	2168	6612	2413	9025
INNs	74	76	66	84	42	108	32	118	150
		50%		56%		72%		79%	
Real data	LOGIT (1)		NB (1)		ANN (0.1905)				Total
	NOINNs	INNs	NOINNs	INNs	NOINNs	INNs			
NOINNs	9025	0	9023	2	7240	1785			9025
INNs	150	0	150	0	53	97			150
		0%		0%		65%			

**Figure 3.** INNs and NOINNs predicted probabilities for the 2013 sample according to the seven machine learning algorithms with SMOTE and the well-behaved example as a benchmark (from the top-left: well-behaved, RPART, TREE, CTREE, BAG, LOGIT, NB, ANN).

as having the best predictive power. In fact, when considering the optimal cut-off in the 2013 sample (0.12), this algorithm classifies 6644 observations as NOINNs and 2531 (38.1%) as INNs. Unfortunately, the predicted probability distributions associated with BAG do not correctly identify many INNs (see

Figure 3), which is not unexpected since it exhibits a bi-modal distribution with a large variance across its domain. We also consider the second-best performing algorithm, ANN, whose distribution of the predicted probability for INNs shows a peak close to 1, although it maintains a second small peak close to 0 (Figure 3).

In order to further increase the performance, instead of using only one algorithm, we consider a weighted mix of BAG and ANN, and we calculate its predicted probability as a convex linear combination of the predicted probabilities originating from the two algorithms independently.⁷ Despite the overall improvement, a large area of common support still remains between INNs and NOINNs densities (Figure 4), and this issue is particularly severe for intermediate values of the predicted probability. Since there is not much difference between the two densities, predictions in that area inevitably lead to a high number of both type I and type II errors. The continuum nature of the empirical problem helps to explain the poor performance of a category classifier. Categories, like ‘innovativeness’, are often the result of an artificial dichotomisation of an otherwise continuous variable: firms can be more or less innovative on a continuum scale. For this reason, when using the model for prediction, instead of introducing only one cut-off, which separates a predicted INN from a predicted NOINN, we identify two cut-offs which identify three intervals in the (0,1) domain of the predicted probability. Firms with a predicted probability smaller than the first cut-off (0.2) are classified as NOINNs while firms with a predicted probability higher than the second cut-off (0.8) are classified as INNs. Unclassified firms, with a predicted probability in-between the two cut-offs, were dropped them from the analysis. The algorithm turns out to perform extremely well in correctly classifying INNs (Table 8): most of the misclassification errors are indeed false negatives. Since this type of error reduces the differences among groups, if we find a difference between innovative and non-innovative firms, the result would hold *a fortiori* with a better algorithm.

4.2.2. Predicting the past: innovative firms in 2008

The mixed BAG-ANN model can now be leveraged to predict which firms would have been classified as innovative in the 2008 sample and to generate a new binary variable (*Inno*). According to Table 9, the sample would remain very unbalanced with 87.8% INNs and 1.9% NOINNs.

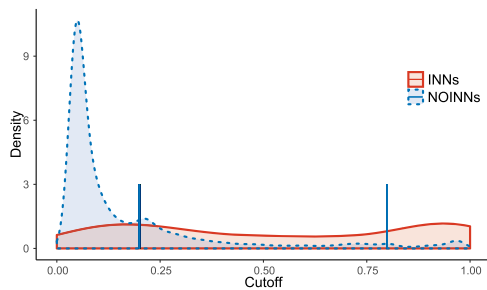


Figure 4. Density of the predicted probabilities for the 2013 sample according to the BAG-ANN mixed model with SMOTE.

Table 8. Predictive performance of the BAG-ANN mixed model on the 2013 sample with different predicted probability cut-offs.

Real data	Prediction					
	cut-off 0.2		cut-off 0.8		Final subsample	
	NOINNs	INNs	NOINNs	INNs	NOINNs	INNs
NOINNs	6751	2274	8698	327	6751	327
	75%		96%		95%	
INNs	40	110	99	51	40	51
		73%		34%		56%

Table 9. BAG-ANN mixed model classification of NOINNs (predicted probability ≤ 0.2) and INNs (predicted probability ≥ 0.8) on the 2008 sample.

	Predicted Probability		Total	%		%
	≤ 0.2	≥ 0.8		≤ 0.2	≥ 0.8	$\geq 0.2 \leq 0.8$
2008	34,487	763	39,295	87.8%	1.9%	10.3 %

We are aware that it is impossible to directly evaluate the performance of the 2008 prediction, and we can only reasonably assume that the true and unknown model which generated the data in 2013 is similar to the one in 2008. Nevertheless, we can provide some statistics on the predicted INNs and NOINNs for a qualitative evaluation of the BAG-ANN performance (Tables 10 and 11). For example, INNs are well represented also in more traditional industries and show significantly higher number of firms involved in R&D and IPR investments and higher average values of investments for the period 2008-2018. The values are significantly higher for the former.

5. Econometric analysis

In this section, we test the hypothesis of a survival premium for 2008 firms classified as INNs with respect to NOINNs.

5.1. Univariate analysis

We first employ the Kaplan-Meier estimator (KME) to show short- and long-term differences in the survival probability of the 2008 firms after the economic and financial crisis.

The KME is a non-parametric estimator classically used to, among other things, estimate survival distribution functions (see, e.g. Fleming and Harrington 1991; Andersen et al. 2012). In general, this analysis studies the closure of start-ups with a survival distribution function $S(t)$: namely, the probability that a start-up will be still alive at time t . Let us consider a sample from the complete set of star-ups with size n (note that here we are dealing with a right-censoring problem). Denote with $t_1 < t_2 < \dots$ the years when start-ups definitely close their activities on the Italian market. Let d_i be the number of start-ups who close at t_i . The Kaplan-Meier estimator $\hat{S}(t)$ for $S(t)$ is:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{r_i}\right), \quad (1)$$

where r_i is the number of start-ups in the risk set just before time t_i , i.e. those firms that had survived, and d_i is the number of closures at time t_i .⁸

The KME allows for direct comparisons across the survival probability of samples with different sizes. Figure 5 shows the two Kaplan-Meier curves with time in years on the horizontal axis and probability of surviving, or proportion of surviving firms, on the vertical axis. The lines represent the survival curves stratified by INNs and NOINNs within their shadows of confidence intervals. At time zero, the survival probability is 1.0 (i.e. 100% of the firms are active). After ten years, the survival probability is approximately 0.687 (or 68.7% - standard deviation 0.002497) for NOINNs and 0.790 (or 79.0% - standard deviation 0.01474) for INNs. INNs enjoy a survival premium and their survival curve always lie above NOINNs.

Table 10. Qualitative evaluation of the mixed model prediction.

	INNs	NOINNs
% of firms with positive R&D investments over 10 years	6%	4%
% of firms with positive IPR investments over 10 years	10%	4%
average R&D investments over 10 years, if positive (€)	612 K	346 K
average IPR investments over 10 years, if positive (€)	7056 K	0,776 K

Note: Differences between groups are statistically significant at the 1% level.

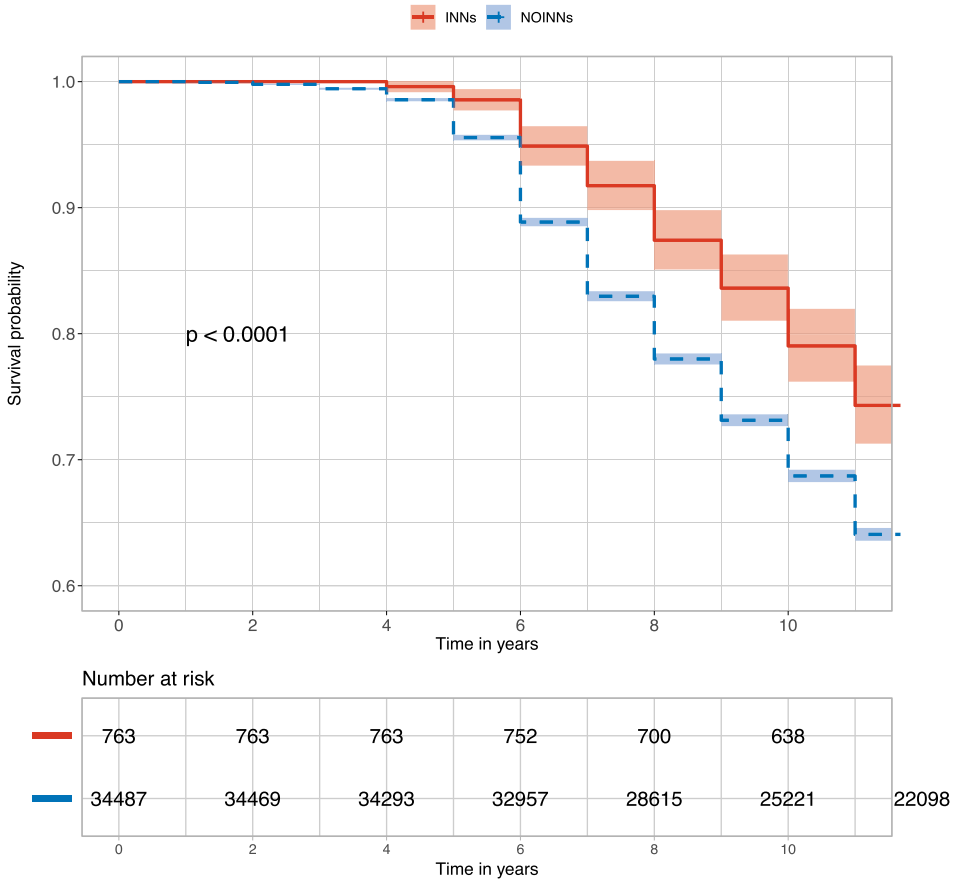


Figure 5. Survival curves of INNs and NOINNs.

Note: the p -value for the log-rank test is reported. Log-minus-log transformation is applied for confidence intervals.

The associated confidence intervals are wider for INNs than for NOINNs, suggesting a higher uncertainty around innovative ventures. Nevertheless, there is always a statistically significant difference in the two groups, as the rejection of the null-hypothesis of the log-rank suggests.

5.2. Multivariate analysis

We now seek to address the last issue raised in the theory, i.e. whether the survival premium of innovative firms persists even when controlling for sectors and locations. We perform this task by adding the one-digit ATECO2007 classification for economic activities, the NUTS3 region classification (namely, 'provincia') for the location effect, and the interaction variables of being INNs with both sector and region as controls in a Cox proportional hazards model. The interaction effect can be interpreted as the positive or negative survival premium linked with the specific sector and region. With the Cox model in Equation (2), we simultaneously estimate the impact of several variables on survival. More precisely, we estimate how the effect of being an INN in a specific sector and in a given location influences the exit rate from the market in a particular year, given that a firm survived up to that year. *Id est*, the hazard rate of failure at time t is;

$$h(t) = h_0(t) \exp\{\beta_1 * Inn + \beta_2 * Sector + \beta_3 * Location + \beta_4 * Interactions\} \quad (2)$$

where:

Table 11. ATECO classification according to the BAG-ANN mixture classification of NOINNs and INNs on the 2008 sample.

ATECO	Predicted ≤ 0.2	Probability ≥ 0.8	Total	ATECO % ≤ 0.2	ATECO % ≥ 0.8
A	743	12	805	92.23%	1.49%
B	41	1	44	93.18%	2.27%
C	3222	109	3903	82.55%	2.79%
D	805	17	880	91.48%	1.93%
E	145	1	170	85.29%	0.58%
F	6936	136	7869	88.14%	1.73%
G	5885	195	7014	93.90%	2.78%
H	946	30	1146	82.55%	2.62%
I	1721	56	2013	85.49%	2.78%
J	1537	37	1754	87.62%	2.11%
K	575	7	627	91.71%	1.12%
L	4763	48	5088	93.36%	0.94%
M	3294	49	3646	90.35%	1.34%
N	1754	32	1982	88.50%	3.85%
O	2	0	5	40.00%	0.00%
P	372	5	401	92.77%	1.25%
Q	594	8	681	87.22%	1.17%
R	689	13	762	90.42%	1.71%
S	423	6	467	90.58%	1.28%
NA	37	1	38	97.37%	2.63%
Total	34,487	763	39,295		

- t is the survival time;
- $h(t)$ is the hazard function;
- β_i are the coefficients. Since the Cox model can be written as a linear regression model of the logarithm of the hazard, it is possible to interpret the $\exp(\beta_i)$ as the hazard ratio of the i^{th} covariate;
- h_0 is the baseline hazard when all the covariates are set equal to zero. It is possible to estimate the β_i without any consideration of the hazard function only under the assumption of proportional hazard, validated both visually and with the log-rank test (see Table 13);
- *Inno*, *Sector* and *Location* are categorical variables summarised in Table 12, while *Interactions* are the interaction terms between *Inno* and the remaining variables.

Table 13 summarises the results of five different models and estimated coefficients. Model (1) uses just the dummy variable for INNs. The coefficient value -0.428 shows that being innovative has a negative and statistically significant effect on the closure probability when compared to NOINNs. A straight interpretation of the effect is to compute the hazard ratio $= e^{-0.428} = 0.65$, i.e. at any given time, innovative firms almost double their chance of survival vis-à-vis NOINNs. Models (2) and (3) add an industrial sector and regional controls, respectively, while models (4) and (5) also consider their interaction effects with INNs. When adding the interaction effects for location, the significance of being innovative disappears. Consistently with economic theory (Feldman 2001), the empirical evidence confirms that a large part of the survival premium depends on location self-selection process, by which any firm, not only INNs, chooses the city or region perceived as the most likely for survival and growth. Yet, this result does not imply that being innovative is irrelevant because being innovative in a specific region can still lead to a survival premium. By looking at the interaction of location with the innovative dummy, we can rank Italian provinces according to the survival premium for being innovative. Figure 6 shows the hazard ratio of the interaction effects when

Table 12. Cox regression: description of the variables.

Name	Description	Categories	Reference	Observation
<i>Inno</i>	Dummy variable for being an INN or a NOINN	2	NOINNs	35,250
<i>Sector</i>	ATECO classification of sectors	20	Manufacturing	35,212
<i>Location</i>	Italian Province (NUTS3 region) in which firm is located	110	Milan	35,250

Table 13. Cox regressions: summary.

	Dependent variable:				
	Failure				
	(1)	(2)	(3)	(4)	(5)
<i>Inno</i>	−0.428*** (0.072)	−0.459*** (0.072)	−0.438*** (0.072)	−0.512*** (0.198)	−0.122 (0.246)
<i>Sector</i>		YES		YES	
<i>Location</i>			YES		YES
<i>Inno * Sector</i>				YES	
<i>Inno * Location</i>					YES
Observations	35,250	35,212	35,250	35,212	35,250
Log Likelihood	−129,172.400	−128,598.200	−128,991.200	−128,591.100	−128,940.600
Wald Test	35.340***	483.320***	381.580***	490.800***	386.730***
LR Test	40.756***	493.846***	403.239***	508.001***	504.494***
Score (Logrank) Test	35.885***	493.376***	388.083***	504.257***	443.604***
Df	1	19	110	36	209

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

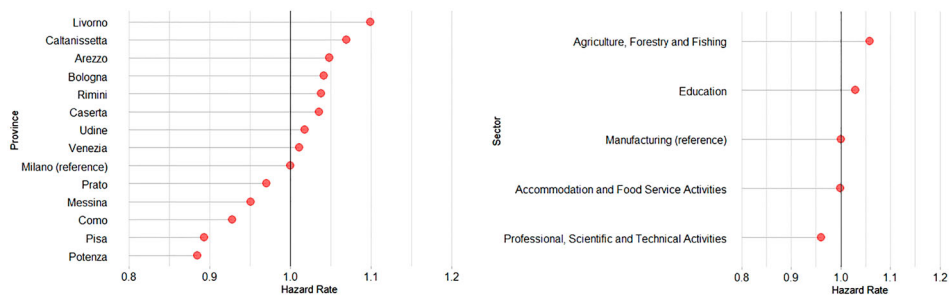


Figure 6. Hazard Ratio for the interaction term (NUTS3 regions and ATECO sector, respectively).

they are statistically significant. The higher the value, the higher the positive effect of innovation on the chance of survival.

It might seem counter-intuitive that the sector control does not absorb the explanation power of INNs, whereas location does. In fact, NUTS3 regions capture a much larger effect, which includes, both the mix of sectors which characterises a geographical area and the dynamics discussed above, such as entrepreneurial atmosphere, agglomeration economies, role of universities, and so forth. Furthermore, we can also compute the magnitude of the interaction effect, also for sectors, as plotted in Figure 6. An inquiry on the causes that make INNs more likely to survive in some locations or sectors is outside the scope of this work, but there is certainly room for new research questions. Note that, at least theoretically, a further model based on the joint estimates of both sectors and locations is possible. Unfortunately, our specification, especially for the 2008 firms classified as INNs, suffers from the complete separation problem (Albert and Anderson 1984), which does not allow for the estimation of certain interaction effects.

6. Conclusion

This paper has contributed to designing a new research framework which combines data science within econometric models. In particular, we used machine learning algorithms to extrapolate information from a large source of data, which could not have been otherwise employed in standard regression models. We stress, and demonstrate, how this exercise needs feedback between economic theory, econometrics and data science, in order to design the desirable properties of the variable created by machine learning algorithms.

We applied this methodological approach to the longstanding debate around the survival premium of innovative start-ups, and employed machine learning to develop a new indicator of innovation at the firm level, which removes some of the drawbacks of previous proxies. During this process, we carefully considered specific constraints to make the new variable suitable for the use in an econometric model. The machine learning algorithm is trained on all possible information (except variables on location and sector, as well as on R&D and IPR). In this way, it was possible to run a survival model which, as suggested by the theory, includes sector and location as *controls* and does not suffer from any form of endogeneity.

This framework can be considered as a weak integration of econometrics and data science since the two approaches are connected via feedback, but they still run separately. Also, it is possible to imagine different scenarios either with a stronger methodological integration, or in which data science completely supersedes econometrics. However, since the state of the art of economics is focused on causal relationships, we believe that such research frameworks have not yet been designed.

As a second contribution, we introduce a new indicator for innovation, suitable for various analyses, since it overcomes many of the main drawbacks of other innovation proxies: it blends different aspects of both the inputs and the outputs of the innovation process. However, its nature is very much connected to the Italian case. In this paper, we use the model to predict the innovativeness of past Italian start-ups, but the same exercise can be done to predict the innovativeness of foreign firms in the present. Indeed, the AIDA Bureau van Dijk database used to train and test the algorithm is consistent with the ORBIS-AMADEUS database, which collects the same information as AIDA on European firms. Nevertheless, the application of a prediction algorithm on a sample in a different country requires considerable efforts in evaluating the results in relation to other measures of innovation, such as patent applications or survey data. However, for the largest European economies, a match between AMADEUS-ORBIS with PATSTAT and CIS data is already in place. Thus, a major line for future work opened up by this paper is the extension of this new measurement to other countries.

As a third contribution, we have provided new empirical evidence on the survival of INNs on the basis of the new indicator. When controlling for sector specificity, INNs seem to maintain their survival premium, which, conversely, fades out when controlling for the location at the NUTS3 regional level. This result challenges the previous literature which formed a weak consensus on the positive effect of innovativeness on survival. We find that INNs have a survival premium only in relation to specific locations. Probably, the specific attributes of a location, which also include the composition of the local economy in term of sectors, might be more or less suitable for a newly established innovative firm. For many locations in the dataset, the effect is not statistically significant, while for some it could not even be estimated due to the small number of start-ups in those areas; however, for other regions, a clear-cut effect exists. Understanding the determinants of survival at the regional level is a question that could be addressed in future work.

Notes

1. Although a stream of literature tries to develop models that overcome the trade-off between the prediction error due to a simple model and the variance of estimates in out-of-sample predictions (Pearl and Mackenzie 2018), in statistical learning, the trade-off is still binding.
2. The debate on the use of patents dates back at least to Pavitt (1985).
3. For a review and future perspectives on history-friendly models see Capone et al. (2019).
4. The 221/2012 Legislative Decree, was adopted, and when in force, on 17 December 2012.
5. See website: <http://startup.registroimprese.it/startup/index.html>
6. Alternatively, as a measure of performance, we can compare the area under the ROC curve (AUC). For further details on the interpretation of ROC curves, see Alpaydin (2014).
7. We respectively assigned the weights 0.77 and 0.23 to BAG and ANN, according to a function which maximises the separation between the predicted probabilities for INNs and NOINNs and the area under the ROC curve (AUC). As a robustness check, we also tested the mix of different algorithms, but there was no substantial improvement in the performance. See Appendix 2 for further details.

8. We estimate the variance with Greenwood's formula using the Delta method, and we use log-minus-log transformation for the confidence interval (Borgan and Liestøl 1990).
9. Note that NAs are much too diffuse among the variables and observations, and therefore multiple imputations will add an extra variability to non justified observed variables. Even if we limit the multiple imputation to some crucial variables, we still do not have enough complete observations in the dataset to finalise the NA completion.
10. Note that management variables, which contain a huge amount of unstandardised text, are discarded from the beginning of the data construction process.
11. Note that, without this last MVA step, there would only have been 18,078 firms left in the 2008 sample, representing less than the 28% of the initial amount of 2008 start-ups.

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No potential conflict of interest was reported by the author(s).

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Appendices

Appendix 1. Missing value analysis

The AIDA database is a valuable source of information, but reporting is mandatory for only a few variables. Hence, a missing value analysis (MVA) is needed to avoid information loss when applying machine learning algorithms, which immediately discards all observations containing missing values (NAs). We propose an MVA to identify variables and observations containing the highest number of NAs and choose which ones to delete. It is a semi-automated approach which balances the loss of information with the introduction of a source of extra variability. No imputation of missing data is undertaken to avoid introducing potential bias in variables with too many NAs.⁹

The MVA starts with the 2013 sample and only afterwards evaluates the status of the 2008 sample. Since, in the 2013 sample, the variables measured in 2015 had not yet been fully incorporated into AIDA at the time of inspection (July 2016), we chose to drop them completely.¹⁰ Hence, we started with 800 available variables (174 do not change over time, while 786 are the results of having observed start-ups for three years: namely, 262×3), and we discarded 262 accounting variables, i.e. we still retained 538 variables for the 68,316 of the 2013 start-ups. We also controlled for the presence of duplicated variables. Subsequently, we defined the number of NAs for each variable and for each firm, obtaining the distributions shown in Table A1 and Figure A1. We observed that the NAs affect both INNs and NOINNs in a similar way. We chose to drop observations with a number of NAs higher than those in the third quartile, i.e. with more than 290 missing over the 538 variables. We obtained a dataset composed of 51,496 observations (including 796 innovative start-ups).

Then, we dropped variables with a number of NAs exceeding those in the first quartile (3968). We kept 51,496 firms observed with 127 variables. In order to employ a model trained on 2013 data for predicting values in 2008, the 2008 and 2013 samples needed to have the same variables. We also made the same analysis on the 2008 sample so as not to lose

Table A1. Missing value distribution observed in 538 variables for the 68,316 observed 2013 start-ups according to the INNs and NOINNs classification.

Missing value	INNs/NOINNs	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Variables	INNs+NOINNs	0	11,170	11,830	19,530	23,620	68,320
	INNs	0.00	88.25	213.00	274.90	297.00	1010.00
	NOINNs	0	11,050	11,620	19,260	23,370	67,310
Firms	INNs+NOINNs	24.0	70.0	98.0	153.8	290.0	530.0
	INNs	32.0	71.0	92.0	146.4	284.0	360.0
	NOINNs	24.0	70.0	98.0	153.9	290.0	530.0

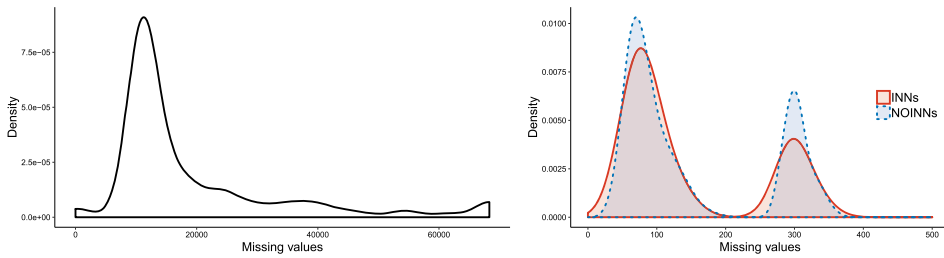


Figure A1. The left panel shows the missing value distribution in 538 variables for the 68,316 observed 2013 start-ups. The right panel shows missing value distributions in observations separated into INNs ($n=1010$) and NOINNs ($n=67,306$), over 538 variables.

too many firms from the 2008 sample. After removing variables already discarded in the 2013 sample, three new variables, containing more than the 30% of the missing values, were identified. Hence, we dropped them from both samples, and we discarded all observations still containing NAs. We obtained two final datasets with 124 variables: the 2013 sample contained 45,576 firms and the 2008 sample contained 39,295 firms.¹¹ The 2008 sample was, then, enriched with further economic variables, such as EBITDA, R&D investments, employees, and IPR investments observed yearly from 2009 to 2018. After the MVA, the proportion of INNs/NOINNs in the 2013 sample become consistent with the original one, growing slightly from 1.5% to 1.59%.

A.1. Algorithms

For the training of the model, we select seven algorithms, known in the machine learning, neural network and econometric literature, briefly presented in what follows.

- (1) The binary recursive partitioning algorithm (RPART) (Breiman et al. 1984) is a tree-based method (Hastie, Tibshirani, and Friedman 2017) grounded in a top-down approach in which the partition starts at the top of the tree. Starting from all observations in a single region, the algorithm only successively splits the space via two further branches of the tree. The Gini's coefficient method is used for the tree variable selection. RPART defines the best split at each step, and predictions are easily interpretable. Differently from other classification algorithms (Gareth et al. 2013), we use RPART in *R* with the `rpart()` function in the `rpart` package.
- (2) The classification tree (TREE) (Breiman et al. 1984; Ripley 1996) is based on binary recursive partitioning, given the classification INNs/NOINNs. It recursively chooses splits from the selected independent variables. Numerical variables are split at a given value α in each node, while categorical variables are split according to two non-empty sets of unordered levels. At each step TREE selects the split which minimises classification impurity. We use TREE in *R* with the `tree()` function in the `tree` package.
- (3) The conditional inference tree (CTREE) (Hothorn, Hornik, and Zeileis 2006; Strasser and Weber 1999) estimates a regression relationship by binary recursive partitioning in a conditional inference framework. It uses a permutation test in order to select the set of variables that maximises the Gini coefficient, differently from other tree-based methods that just select one variable at each step. We use CTREE in *R* with the `ctree()` function in the `party` package.
- (4) The bagging algorithm (BAG) (Breiman et al. 1984; Strobl, Malley, and Tutz 2009; Gareth et al. 2013), or bootstrap aggregation, is based on the necessity to reduce the variance of the statistical learning tree previously described. It is based simply on the idea that the variance can be reduced if, instead of only one training set, we use the average of more training sets. For this reason, it is based on the aggregation of many decision trees. BAG generates M different bootstrapped training data sets (with an increment in computation time), and then it trains the method on the M bootstrapped sets in order to average all the obtained predictions. Here we use RPART as the basis of BAG. We use BAG in *R* with the `bagging()` function in the `ipred` package.
- (5) The Logit regression model (LOGIT) (McCullagh and Nelder 1989) is used here as the benchmark and widely-used econometric model. It can be seen as a generalised linear model based on a Logit link function (Agresti 2002; McCullagh and Nelder 1989) or a random utility model for discrete choices (Train 2009). It estimates a linear relation between the independent variables and the Logit of the INNs probability. Its accuracy suffers in the presence of huge datasets, as in our case (Perlich, Provost, and Simonoff 2003). We use LOGIT in *R* with the `glm()` function.
- (6) The naïve Bayesian classifier (NB) (Alpaydin 2014), in its particular binary version, is based on the estimation of the conditional a-posterior probabilities of INNs, given the selected independent predictors, using the well-known Bayes rule. We use NB in *R* with the `naiveBayes()` function in the `e1071` package.
- (7) The artificial neural network (ANN) (Bishop 1995; Ripley 1996) is a single hidden layer back-propagation network (Hastie, Tibshirani, and Friedman 2017). It is based on the artificial reproduction of the functioning of the brain (Posner 1989), therefore ANN is a nonlinear statistical model based on a two-stage estimator. We use ANN in *R* with the `nnet()` function in the `nnet` package.

Appendix 2. Optimal cut-off and mixture weight optimisation

Part of the methodology introduced in this contribution is new, therefore new *R* functions have been coded to undertake the analysis. A first built-in function implements three criteria for the selection of the optimal cut-off in each algorithm:

- (1) the Youden index (*J*) method, which defines the optimal cut-point as the point maximising the difference between the true positive rate and the false positive rate (namely, the Youden function) over all possible cut-point values;
- (2) the point closest to the (0,1) corner in the ROC plane method, which defines the optimal cut-point as the point which minimises the Euclidean distance between the ROC curve and the (0,1) point;
- (3) the optimal cut-point method which selects as the optimal cut-off, the point which maximises the product of sensitivity and specificity.

The confusion matrix in [Table 7](#) is the result of the application of the second criterion. Similar results have been obtained applying the other two approaches.

A second built-in *R* function finds the optimal mixture weights, following the approach described below. First, we select two (or more) candidate algorithms to compose the mixture (here *alg1* =BAG and *alg2* =ANN), according to their performance emerging from the study of the ROC curve and of the confusion matrix. Second, we retain the predicted probabilities (*pred.prod*), under the selected algorithms, for INNs (positive) and NOINNs (negative). Third, we select a mixture of weights α and $1 - \alpha$ in the support (0,1) according to an optimisation process. The latter simultaneously maximises, for all α in the support, (i) the Euclidean distance between the INNs and the NOINNs predicted probabilities; and (ii) the area under the ROC (AUC). A unique solution of this maximisation process exists and selects α , such that the predicted probability of the mixture is defined as follows:

$$pred.prob.mixture = \alpha * pred.prod.alg1 + (1 - \alpha) * pred.prod.alg2.$$

Appendix 3. Further descriptive statistics on the data

Further descriptive statistics on the 2013 sample are here proposed. We study the distribution of employments ([Figure A2](#)) and EBITDA ([Figure A3](#) and [Table A4](#)) over the two first years of activity (2013 and 2014) according to the type of start-ups registered in the special section of the Italian companies register. We also compare 2013 geographical distributions, at NUTS3 level, of INNs and NOINNs (see [Figure A4](#)).

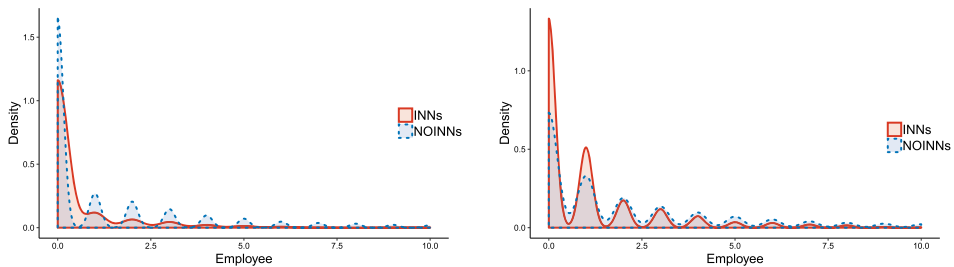


Figure A2. Employees distributions in 2013 and 2014 of INNs and NOINNs in the 2013 sample.

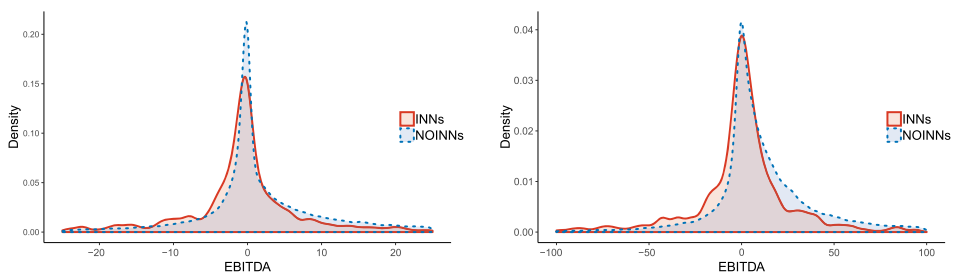


Figure A3. EBITDA distributions in 2013 and 2014 of INNs and NOINNs in the 2013 sample.

Table A2. Observations in the starting and final samples both in 2008 and 2013.

Sample	2008	2013
Total	65,088	68,316
NA	25,793	22,740
Final sample	39,295	45,576

Table A3. Juridical form of the 2013 innovative start-ups ($n=1010$).

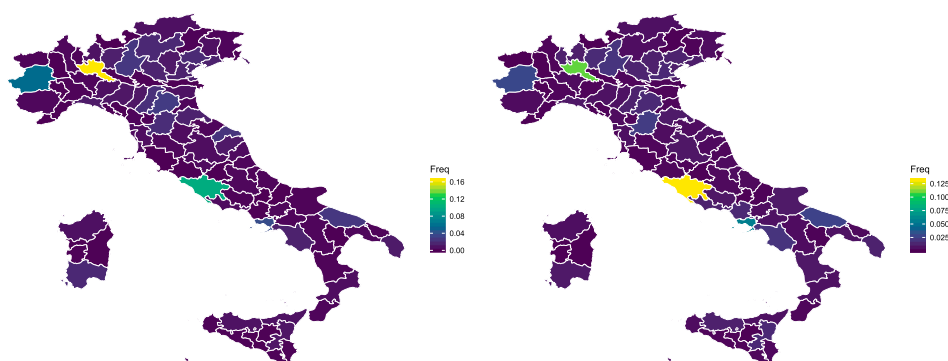
Juridical form	Number of innovative start-ups
S.C.A.R.L.P.A.	14 (1.39%)
S.P.A.	13 (1.29%)
S.R.L.	820 (81.19%)
S.R.L. a capitale ridotto	11 (1.09%)
S.R.L. a socio unico	45 (4.46%)
S.R.L. semplificata	106 (10.50%)
Società consortile a responsabilità limitata	1 (0.1%)

Source: AIDA.

Table A4. 2013 innovative start-ups - EBITDA distribution observed in the first (2013) and the second (2014) year of activity.

Var		Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's	<i>t</i> -stat	<i>p</i> -val
EBITDA	INNs	−537.40	−3.68	−0.39	−3.02	1.95	207.20	213 (21.09%)	4.9949	0.00
2013	NOINNs	−12360	−1.53	0.01	11.73	7.51	120,000	9946 (14.78%)		
EBITDA	INNs	−895.20	−10.22	0.09	−11.37	9.18	379.40	78 (7.72%)	7.7044	0.00
2014	NOINNs	−18090	−1.18	6.43	37.66	25.43	261,000	11,110 (16.51%)		

Source: AIDA.

**Figure A4.** Geo-localisation of the 2013 sample. The INNs are presented in the left panel, and the NOINNs in the right one.